

Exercise 1: Singleton Pattern in C#

Project Creation Commands

```
cd "C:\Users\vanka\OneDrive\Desktop\Design_Patterns_"
```

```
mkdir SingletonPattern
```

```
cd SingletonPattern
```

```
dotnet new console
```

```
code .
```

File Name: **Logger.cs**

```
public sealed class Logger
{
    private static Logger instance;

    private Logger()
    {
    }

    public static Logger GetInstance()
    {
        if (instance == null)
        {
            instance = new Logger();
        }
        return instance;
    }

    public void Log(string message)
```

```
{  
    Console.WriteLine("LOG: " + message);  
}  
}
```

File Name: Program.cs

```
Logger logger1 = Logger.GetInstance();
```

```
Logger logger2 = Logger.GetInstance();
```

```
logger2.Log("Logging Information");
```

```
if (logger1 == logger2)
```

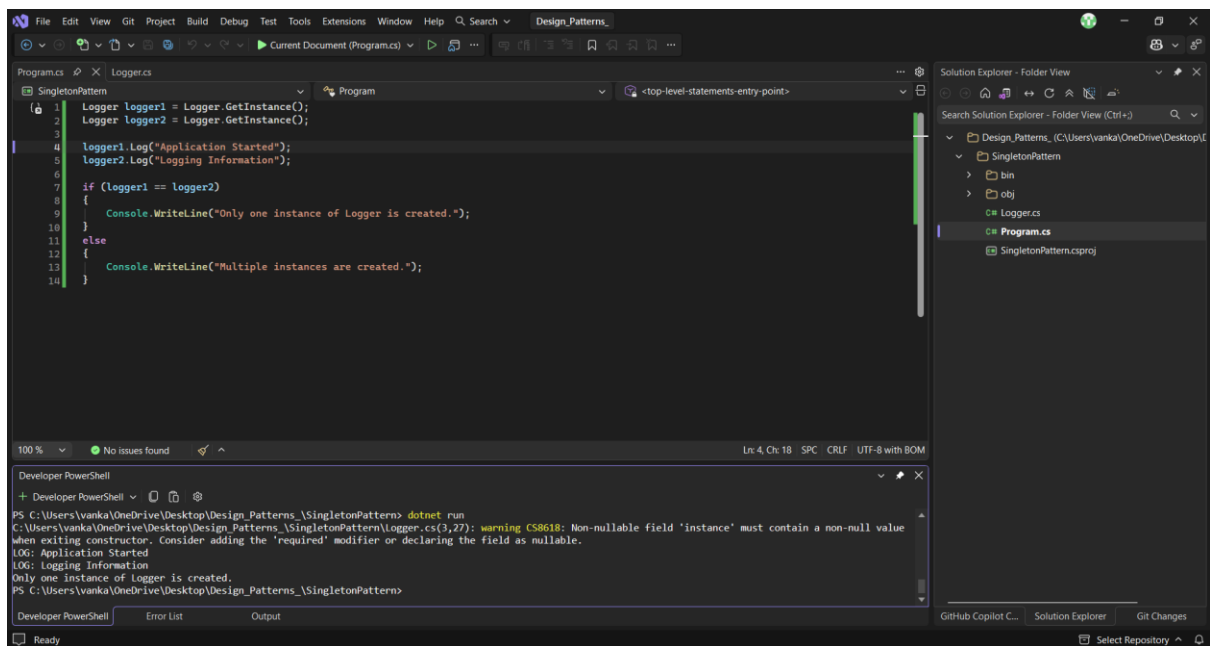
```
{  
    Console.WriteLine("Only one instance of Logger is created.");  
}
```

```
else
```

```
{  
    Console.WriteLine("Multiple instances are created.");  
}
```

Run Command

```
dotnet run
```



Expected Output

LOG: Application Started

LOG: Logging Information

Only one instance of Logger is created.